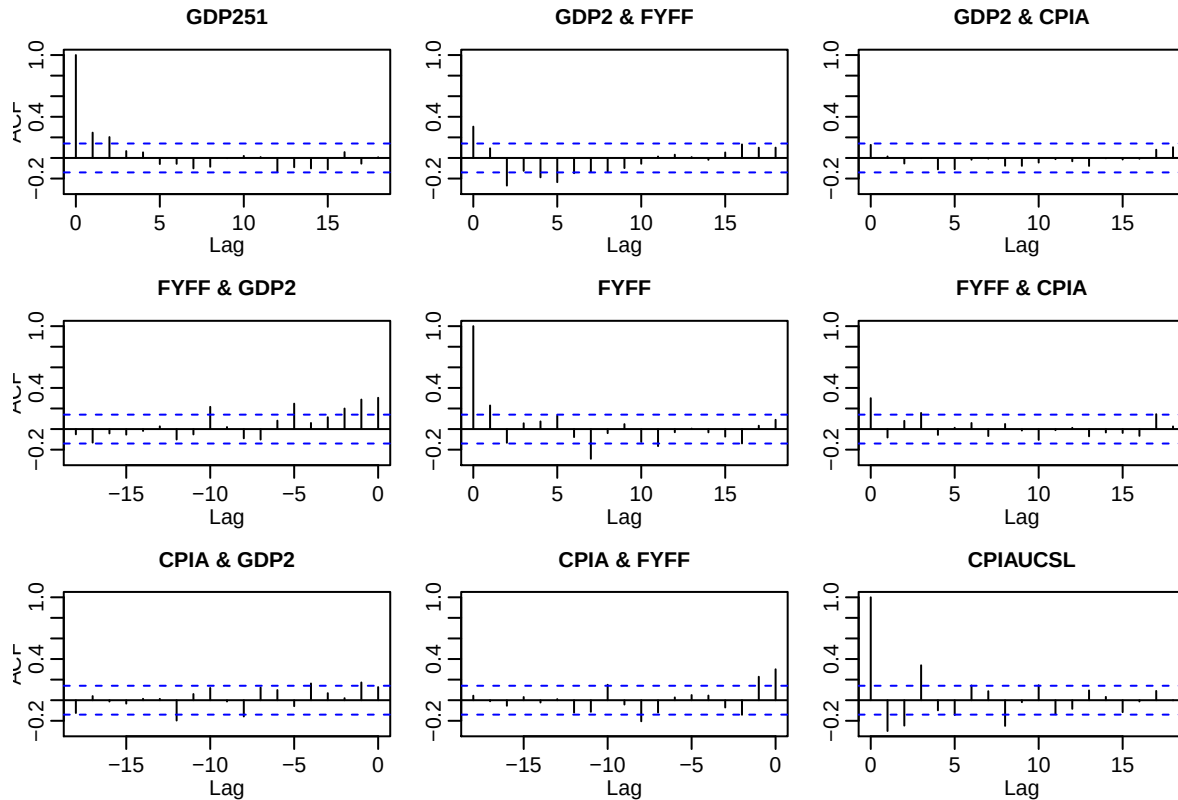


HW5 – Ying Zheng

Problem 1

(a)

```
data= read.table("MacroVars.csv",sep=",", header=T)
acf(data)
```



Comments: The plots above showed us acf and ccf of the 3 macro variables GDP, FFR and CPI. On the diagonal are acfs of them, and the rest are ccfs. We notice that GDP is correlated with FFR at lag 0, -1, -5 and -10, while it has barely any correlation with CPI at all lags. FFR has barely any correlation with CPI at all lags either.

(b)

```
holdfit = VAR(data, lag.max=20)
holdfit$p
```

```
## AIC(n)
##      10
```

(c)

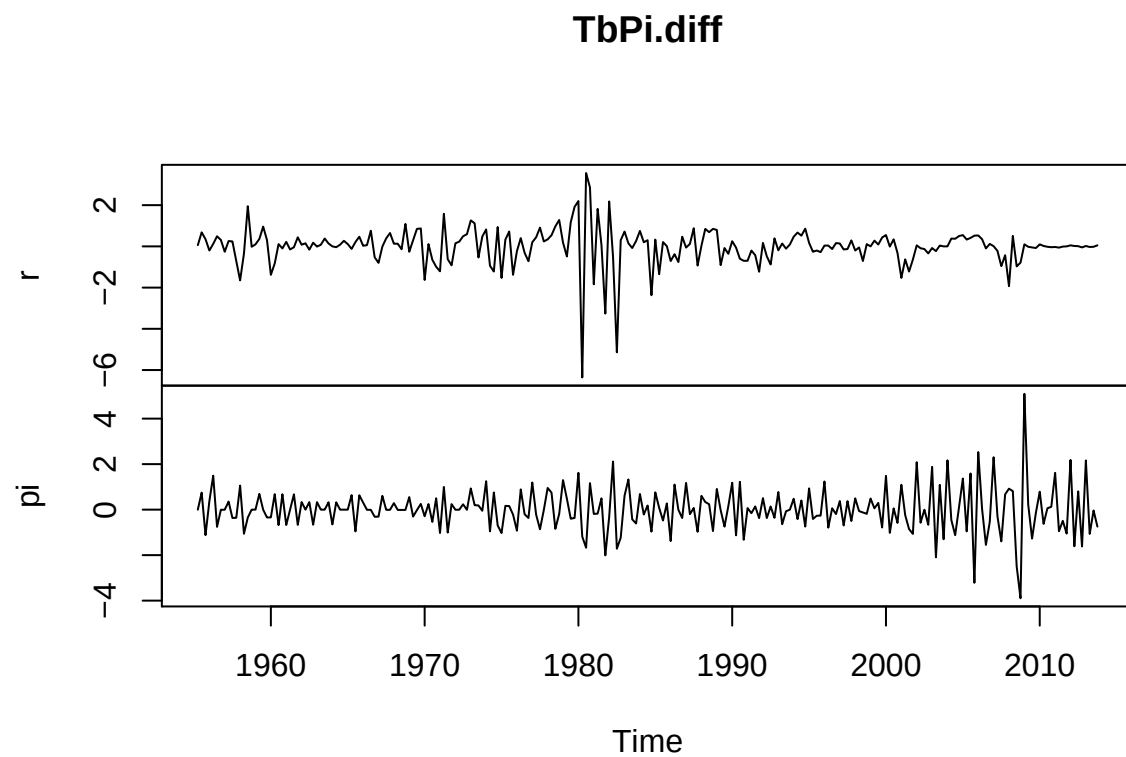
```
serial.test(holdfit)
```

```
##
## Portmanteau Test (asymptotic)
##
## data: Residuals of VAR object holdfit
## Chi-squared = 76.129, df = 54, p-value = 0.02528
```

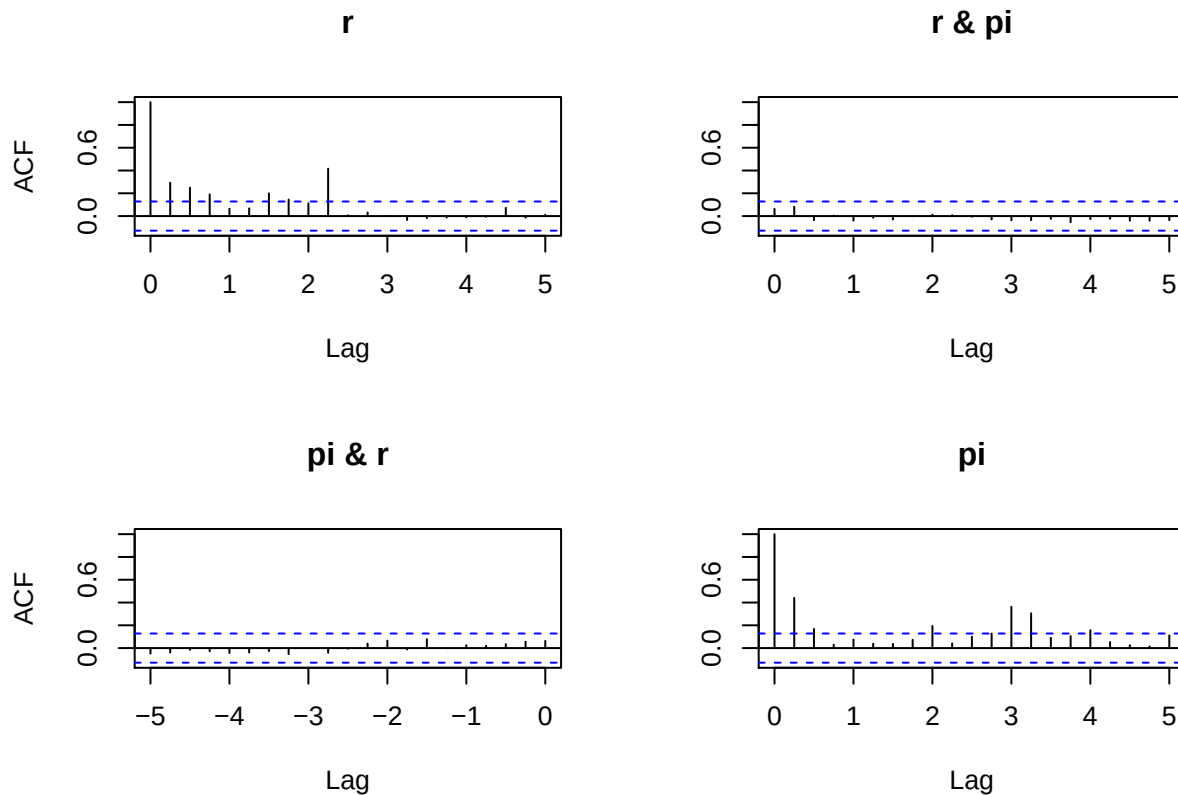
The p-value is 0.02528, which is smaller than 0.05, shows evidence of serial correlation in residuals.

Problem 2 (a)

```
TbGdpPi = read.csv("TbGdpPi.csv", header=TRUE)
TbPi.diff = ts(apply(TbGdpPi[, -2], 2, diff), start=c(1955, 2), freq=4)
plot(TbPi.diff)
```



```
acf(TbPi.diff~2)
```



```
source("SDAFE2.R")
```

```
## Loading required package: fGarch
## Loading required package: timeDate
## Loading required package: timeSeries
##
## Attaching package: 'timeSeries'
## The following object is masked from 'package:zoo':
##
##   time<-
## Loading required package: fBasics
```

```
mLjungBox(TbPi.diff^2, lag=8)
```

```
##   K   Q(K) d.f. p-value
## 1 8 132.57  32      0
```

Yes. First, the plots above shows clustering effects. Secondly, from the Ljung-Box test, there is strong evidence of correlation in the joint series. Finally, from the plot of TbPi.diff, we can see unstationarity. So it exhibit conditional heteroskedasticity.

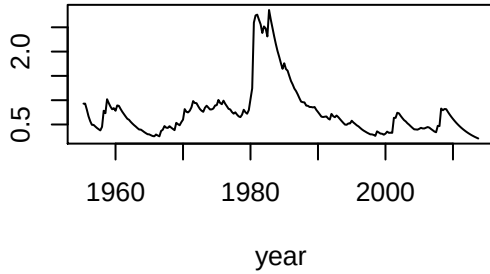
(b)

```
EWMA.param = est.ewma(lambda.0=0.95, innov=TbPi.diff)
EWMA.param$lambda.hat
```

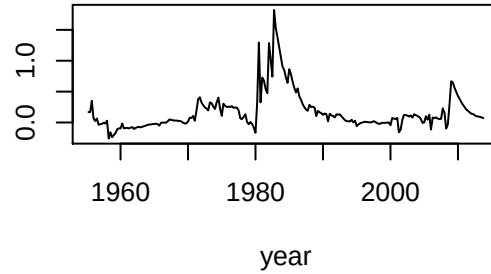
```
## [1] 0.8665812
```

```
EWMA.Sigma=sigma.ewma(lambda=EWMA.param$lambda.hat,innov=TbPi.diff)
par(mfrow = c(2,2))
plot(ts(EWMA.Sigma[1,1]^0.5, start = c(1955, 2), frequency = 4),
     type = 'l', xlab = "year", ylab = NULL,
     main = expression(paste("(a) ", hat(sigma)["1,t"])))
plot(ts(EWMA.Sigma[1,2], start = c(1955, 2), frequency = 4),
     type = 'l', xlab = "year", ylab = NULL,
     main = expression(paste("(b) ", hat(sigma)["12,t"])))
plot(ts(EWMA.Sigma[1,2]/(sqrt(EWMA.Sigma[1,1]* EWMA.Sigma[2,2])),
     start = c(1955, 2), frequency = 4),
     type = 'l', xlab = "year", ylab = NULL,
     main = expression(paste("(c) ", hat(rho)["12,t"])))
plot(ts(EWMA.Sigma[2,2]^0.5, start = c(1955, 2), frequency = 4),
     type = 'l', xlab = "year", ylab = NULL,
     main = expression(paste("(d) ", hat(sigma)["2,t"])))
```

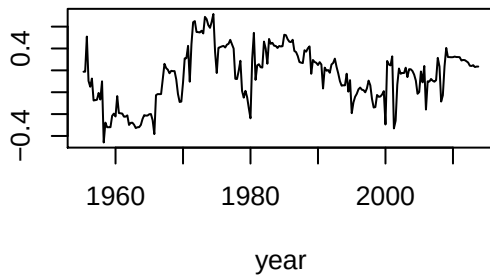
(a) $\hat{\sigma}_{1,t}$



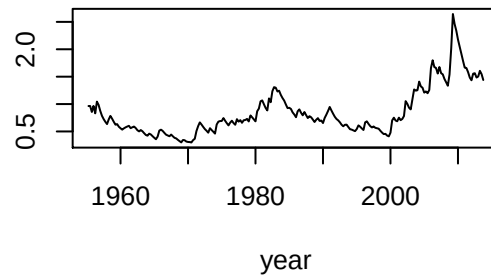
(b) $\hat{\sigma}_{12,t}$



(c) $\hat{\rho}_{12,t}$



(d) $\hat{\sigma}_{2,t}$



From the estimates above, the estimated parameter is 0.8665812.

(c)

```
n = dim(TbPi.diff)[1]
d = dim(TbPi.diff)[2]
stdResid.EWMA = matrix(0,n,d)
for(t in 1:n){
  stdResid.EWMA[t,] = TbPi.diff[t,] %*% matrix.sqrt.inv(EWMA.Sigma[,t])
}
mLjungBox(stdResid.EWMA^2, lag=8)
```

```
## K Q(K) d.f. p-value
## 1 8 60.22 32 0.002
```

Form the Ljung-Box test of the squared standardized residuals, the p-value is 0.002, which is significantly smaller than 0.05, showing strong evidence of serial correlation. So the EWMA model is not adequate.